Achieve!

Surface area and reaction rates

Shelby conducted an experiment to investigate the effect of surface area on the reaction rate between hydrochloric acid and magnesium. She compared powdered magnesium to a piece of magnesium with the same mass. She placed the conical flask containing the magnesium and acid on electronic scales and recorded the loss of mass over a period of time. When there was no longer a loss in mass, she knew the reaction had finished.

☞ **1** Listed below are the two reactants and two products for this experiment. Decide which are which, and then fill in the word equation.

hydrogen gas	hydrochloric acid	magnesium chloride	magnesium

_____ + _____ ⟶ _____ + _____

2 Which product do you think was able to escape the conical flask? _____

Shelby's results are recorded below.

	0 min	**1 min**	**2 min**	**3 min**	**4 min**	**5 min**	**6 min**	**7min**	**8 min**	**9 min**
powdered magnesium (loss of mass, g)	0	0.4	0.7	0.8	0.85	0.9	0.9	0.9	0.9	0.9
piece of magnesium	0	0.25	0.3	0.45	0.6	0.7	0.75	0.85	0.9	0.9

3 Draw a line graph of her results in your workbook. Draw one line for powdered magnesium and one line for the piece of magnesium. Include a key.

4 Which type of magnesium has the fastest rate of reaction? _____

5 When did the reaction finish for the:

a powder? _____

b piece of magnesium? _____

6 Why do you think this happened? _____

_____

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